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SQL QURIES:

21-08-2026

DROP DATABASE IF EXISTS joins_practice;

CREATE DATABASE joins_practice;

USE joins_practice;

CREATE TABLE departments (

dept_id INT PRIMARY KEY,

dept_name VARCHAR(30),

location VARCHAR(30)

);

CREATE TABLE employees (

emp_id INT PRIMARY KEY,

emp_name VARCHAR(30),

dept_id INT,

manager_id INT,

salary DECIMAL(10,2),

hire_date DATE,

FOREIGN KEY (dept_id) REFERENCES departments(dept_id)

);

CREATE TABLE projects (

project_id INT PRIMARY KEY,

project_name VARCHAR(50),

dept_id INT,

budget DECIMAL(12,2),

FOREIGN KEY (dept_id) REFERENCES departments(dept_id)

);

```
CREATE TABLE employee_project (  
    emp_id INT,  
    project_id INT,  
    hours INT,  
    PRIMARY KEY (emp_id, project_id),  
    FOREIGN KEY (emp_id) REFERENCES employees(emp_id),  
    FOREIGN KEY (project_id) REFERENCES projects(project_id)  
);
```

```
INSERT INTO departments VALUES
```

```
(10, 'Engineering', 'Chennai'),  
(20, 'Sales', 'Hyderabad'),  
(30, 'HR', 'Bengaluru'),  
(40, 'Research', 'Pune'),  
(50, 'Finance', 'Mumbai');
```

```
INSERT INTO employees VALUES
```

```
(101, 'Arjun', 10, NULL, 90000, '2021-01-15'),  
(102, 'Priya', 10, 101, 65000, '2022-03-10'),  
(103, 'Karthik', 20, 106, 55000, '2022-07-20'),  
(104, 'Meena', 30, NULL, 60000, '2020-11-05'),  
(105, 'Rahul', NULL, 101, 50000, '2023-02-12'),  
(106, 'Divya', 20, NULL, 85000, '2019-06-25'),  
(107, 'Sanjay', 10, 101, 72000, '2021-09-17'),  
(108, 'Anitha', 40, NULL, 78000, '2020-04-30'),  
(109, 'Vikram', 20, 106, 48000, '2024-01-08');
```

```
INSERT INTO projects VALUES
```

```
(201, 'Banking App', 10, 500000),  
(202, 'Sales Portal', 20, 300000),  
(203, 'AI Research', 40, 800000),
```

```
(204, 'Employee System', 30, 250000),  
(205, 'Cloud Migration', 10, 600000),  
(206, 'Audit Automation', 50, 350000);
```

```
INSERT INTO employee_project VALUES
```

```
(101, 201, 120),  
(102, 201, 100),  
(102, 205, 80),  
(103, 202, 110),  
(106, 202, 70),  
(107, 201, 90),  
(107, 205, 130),  
(108, 203, 150),  
(104, 204, 60);
```

```
SELECT * FROM departments;
```

```
SELECT * FROM employees;
```

```
SELECT * FROM projects;
```

```
SELECT * FROM employee_project;
```

```
SELECT e.emp_id, e.emp_name, d.dept_name, d.location FROM employees e  
INNER JOIN departments d ON e.dept_id = d.dept_id;
```

```
SELECT e.emp_id, e.emp_name, d.dept_name  
FROM employees e LEFT JOIN departments d ON e.dept_id = d.dept_id;
```

```
SELECT d.dept_id, d.dept_name  
FROM departments d LEFT JOIN employees e ON d.dept_id = e.dept_id WHERE e.emp_id IS NULL;
```

```
SELECT e.emp_name, p.project_name, d.dept_name, ep.hours  
FROM employees e
```

```
JOIN employee_project ep
ON e.emp_id = ep.emp_id
JOIN projects p
ON ep.project_id = p.project_id
JOIN departments d
ON p.dept_id = d.dept_id;
```

```
SELECT p.project_id, p.project_name, e.emp_name
FROM projects p
LEFT JOIN employee_project ep
ON p.project_id = ep.project_id
LEFT JOIN employees e
ON ep.emp_id = e.emp_id;
```

```
SELECT e.emp_name AS employee, m.emp_name AS manager
FROM employees e
LEFT JOIN employees m
ON e.manager_id = m.emp_id;
```

```
SELECT e.emp_id, e.emp_name, COUNT(ep.project_id) AS total_projects
FROM employees e
JOIN employee_project ep
ON e.emp_id = ep.emp_id
GROUP BY e.emp_id, e.emp_name
HAVING COUNT(ep.project_id) > 1;
```

```
SELECT d.dept_name, COUNT(e.emp_id) AS employee_count, COALESCE(AVG(e.salary), 0) AS
average_salary
FROM departments d
LEFT JOIN employees e
```

```
ON d.dept_id = e.dept_id  
GROUP BY d.dept_id, d.dept_name;
```

```
SELECT e.emp_name,  
       ed.dept_name AS employee_department,  
       p.project_name,  
       pd.dept_name AS project_department  
FROM employees e  
JOIN employee_project ep  
ON e.emp_id = ep.emp_id  
JOIN projects p  
ON ep.project_id = p.project_id  
JOIN departments ed  
ON e.dept_id = ed.dept_id  
JOIN departments pd  
ON p.dept_id = pd.dept_id  
WHERE e.dept_id <> p.dept_id;
```

```
SELECT p.project_id,  
       p.project_name,  
       d.dept_name,  
       p.budget  
FROM projects p  
JOIN departments d  
ON p.dept_id = d.dept_id  
LEFT JOIN employee_project ep  
ON p.project_id = ep.project_id  
WHERE ep.emp_id IS NULL;
```